Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Options and Futures

Song Xiao

xiaosong1@cuhk.edu.cn

Fall Term 2025

© S.Xiao (2025)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
▷ Forwards and Futures
Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Forwards and Futures

Outline

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
▷ Forwards and Futures
Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Forward contracts
- Futures

Forward contracts

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- A forward contract entered into at time t with delivery date T and forward price F_t is a commitment to buy/sell the underlying at price F_t at time T
  - The parties to the contract agree on the price F_t at the initiation of the contract.
  - Payment and delivery take place at the future date T.
- The party who buys forward is said to be long in the contract.
- The party who sells forward is said to be short in the contract.

© S.Xiao (2025)

Net cash flows for a forward contract

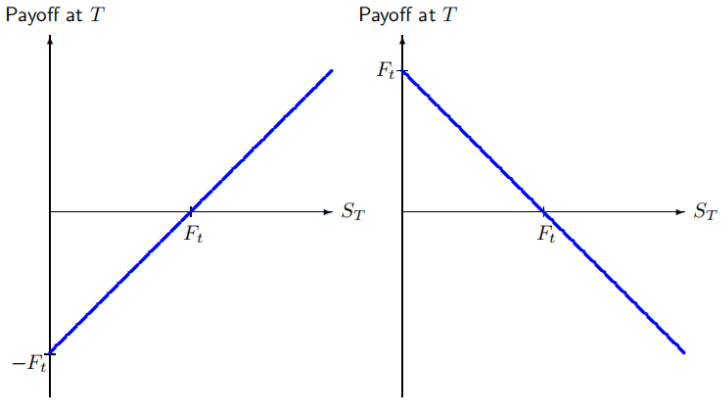
Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Let S_T denote the price of the underlying at maturity
  - Payoff at T on the long side of the contract is $S_T - F_t$
  - Payoff at T on the short side of the contract is $F_t - S_T$

  Note: The payoff is a linear function of S_T.
- At the initiation of the contract, there is no exchange of cash: A forward position is entered at no cost.
- At time t, F_t has to be set such that both parties agree to enter into the contract.
- No-arbitrage arguments allow us to pin down F_t.

Payoff of a long/short position in a forward contract

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



Pricing forward contracts

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Pricing a forward contract means finding F_t such that the possibility of going long or short in the contract (at no initial cost) at this specified forward price does not create an arbitrage opportunity.
- We will assume that there is a constant riskfree rate r (continuously compounded). We will drop this assumption later.

© S.Xiao (2025)

No-arbitrage forward price

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- **Proposition 1**: The forward price for a non-dividend paying asset is

$$F_t = e^{r(T-t)} S_t.$$

- Proof:
  - If you buy the asset at price S_t and sell it forward at the forward price F_t (i.e. short one forward contract), it costs you S_t initially and you have a deterministic payoff F_t at maturity.
  - Because your payoff is riskless, absence of arbitrage implies that you earn a rate of return equal to the riskfree rate, i.e.

$$F_t = e^{r(T-t)} S_t.$$

Alternative proof (by replication)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- If you buy forward, your payoff at maturity is $S_T - F_t$.
- Consider the following static strategy: At time t, buy the underlying and borrow $e^{-r(T-t)}F_t$ with repayment at time T.
- This strategy has the same payoff as the long forward position, at an initial cost of $S_t - e^{-r(T-t)}F_t$.
- To prevent arbitrage, it must be that $S_t - e^{-r(T-t)}F_t = 0$, i.e.

$$F_t = e^{r(T-t)}S_t.$$

Alternative proof II (risk-neutral valuation)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Suppose you take a long position in a forward contract at t. You pay nothing at t and receive $S_T - F_t$ at T.
- By the FTAP,

$$0 = e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}(S_T - F_t)$$
$$\Rightarrow 0 = \mathbb{E}_t^{\mathbb{Q}}(S_T - F_t).$$

- It follows that

$$F_t = \mathbb{E}_t^{\mathbb{Q}}(S_T).$$

- Applying the FTAP to the price of the underlying,

$$S_t = e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}(S_T).$$

- Hence,

$$F_t = e^{r(T-t)}S_t.$$

Comment on the formula $F_t = e^{r(T-t)}S_t$

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- $\mathbb{E}_t(S_T)$ does not enter the formula.
- When $r > 0$, $F_t > S_t$. This makes sense because whether you buy the asset spot or forward, you hold the asset at time T but when you buy it forward you can save at rate r in between now and time T.
  – i.e. buying spot has an opportunity cost.
- Note: The latter argument does not carry through if the asset pays dividends or brings some other kind of benefit while you hold it between t and T.
  – In that case, buying the asset forward as opposed to spot becomes relatively less appealing, which has to be reflected in a lower forward price.

No-arbitrage forward price with dividends

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- **Proposition 2**: Consider an asset which pays at time $\tau \in (t, T)$ a deterministic lump-sum dividend with present value I_t as of time t. Its forward price is
$$F_t = e^{r(T-t)}(S_t - I_t).$$
- Proof:
  – If you buy the asset at price S_t and sell it forward at the forward price F_t, you get two riskless cash flows: $e^{r(\tau-t)}I_t$ at τ, and F_t at T.
  – Absence of arbitrage implies that S_t is equal to the present value of these cash flows, i.e. $S_t = I_t + e^{-r(T-t)}F_t$.
  – Hence $F_t = e^{r(T-t)}(S_t - I_t)$.
- Note: $S_t - I_t$ is the price of the asset which is the real object of the forward transaction, i.e. a claim to cash flows from date T onwards.

Alternative argument (by replication)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The portfolio which replicates the payoff of a long forward position involves buying the underlying, borrowing $e^{-r(T-t)}F_t$ with repayment at time T and borrowing I_t with repayment at time τ.
- This portfolio has a payoff $S_T - F_t$ at time T, involves no net cash flow at time τ and has an initial cost of $S_t - e^{-r(T-t)}F_t - I_t$.
- Absence of arbitrage requires that $S_t - e^{-r(T-t)}F_t - I_t = 0$, which is equivalent to $F_t = e^{r(T-t)}(S_t - I_t)$.

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- **Proposition 3**: The forward price for an asset paying a constant dividend yield q between dates 0 and T is

$$F_t = e^{(r-q)(T-t)} S_t.$$

- Recall that saying that an asset (e.g. a stock index) pays a constant dividend yield q means that at every instant t it pays an instantaneous dividend given by $dD_t = qS_t dt$.
- Let H_t denote your holdings of the underlying asset at time t. If you reinvest the instantaneous dividend in the asset, your holdings of the asset increase by

$$dH_t = \frac{H_t \times qS_t dt}{S_t} = qH_t dt.$$

- Therefore, if you start with H_t at time t and keep reinvesting the dividends between times t and T, you end up with

$$H_T = e^{q(T-t)} H_t.$$

© S.Xiao (2025)

Proof of Proposition 3

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Suppose you buy one unit of the asset at price S_t at time t and keep reinvesting the dividends until the maturity date T. You will end up with $e^{q(T-t)}$ units of the asset at T.
- Therefore, by selling $e^{q(T-t)}$ units forward at price F_t, you can lock in a riskless final payoff $e^{q(T-t)}F_t$.
- Since your initial cost was S_t, absence of arbitrage requires that $e^{q(T-t)}F_t = e^{r(T-t)}S_t$, which is equivalent to $F_t = e^{(r-q)(T-t)}S_t$.
- Alternatively, you can replicate the date T payoff $S_T - F_t$ by buying $e^{-q(T-t)}$ units of the asset, borrowing the present value of F_t, and reinvesting the dividends until T. The cost of this strategy is $e^{-q(T-t)}S_t - e^{-r(T-t)}F_t$, which must be equal to zero. The result follows.

Particular case: Forward exchange rate

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Take the particular case where the underlying is a foreign currency and assume an instantaneous riskfree rate of return r^* is available in the foreign country.
- i.e. the foreign currency earns a constant dividend yield equal to r^*.
- Let S denote the price in domestic currency of one unit of foreign currency. Thus S_t is the current exchange rate. The no-arbitrage value of the forward exchange rate is

$$F_t = e^{(r-r^*)(T-t)} S_t.$$

- This is the so-called covered interest rate parity.
- Remark: This is equivalent to $e^{r(T-t)} = e^{r^*(T-t)} \frac{F_t}{S_t}$, which is easily interpreted as a no-arbitrage condition.

Storage costs for commodities

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- When the underlying is a commodity, storage costs become relevant. These costs make buying forward rather than spot more appealing because it allows you to save these costs – this should be reflected in a higher forward price.
- Storage costs can be treated as negative dividends.
- **Proposition 4**: Consider a forward contract whose underlying is a commodity. If the present value of storage costs over the life of the contract is U_t, the forward price of the commodity is

$$F_t = e^{r(T-t)}(S_t + U_t).$$

- In the particular case where storage costs at any time are proportional to the price of the commodity (instantaneous cost at time t equal to $uS_t dt$), the forward price of the commodity is

$$F_t = e^{(r+u)(T-t)}S_t.$$

Summary: Cost of carry formula

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- The relationship between the current underlying spot price and the forward price can be summarized as:

$$F_t = e^{(r+u-q)(T-t)} S_t = e^{c(T-t)} S_t,$$

  where $c := r + u - q$ is called the cost of carry.
- The interest rate r is an opportunity cost, u a storage cost, and $-q$ is a negative cost corresponding to the dividend yield.
  – The forward curve is the term structure of forward prices (F_t as a function of T).
- If $c > 0$, $F_t > S_t$, and the forward curve is upward sloping. This is called a normal market.
- If $c < 0$, $F_t < S_t$, and the forward curve is downward sloping. This is called an inverted market.

Convenience yield

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- For some commodities, it might be that some benefits are derived from owning/holding the underlying.
  – Then buying spot is relatively more appealing than buying forward, which should be reflected in a lower forward price:

$$F_t < e^{c(T-t)} S_t.$$

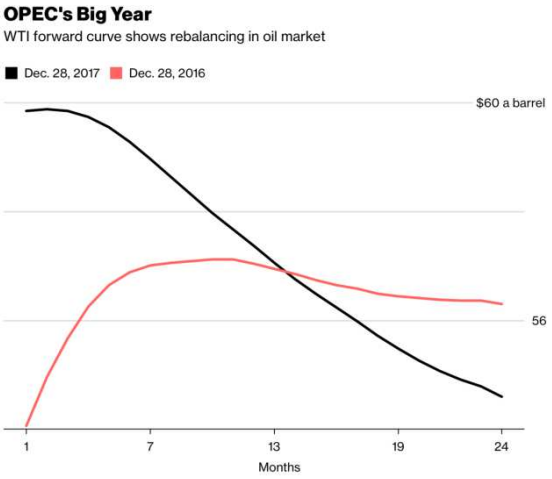
- The convenience yield y on a commodity is such that:

$$F_t = e^{(r-y)(T-t)} S_t.$$

  – Note: convenience yield and dividend yield play in the same way.

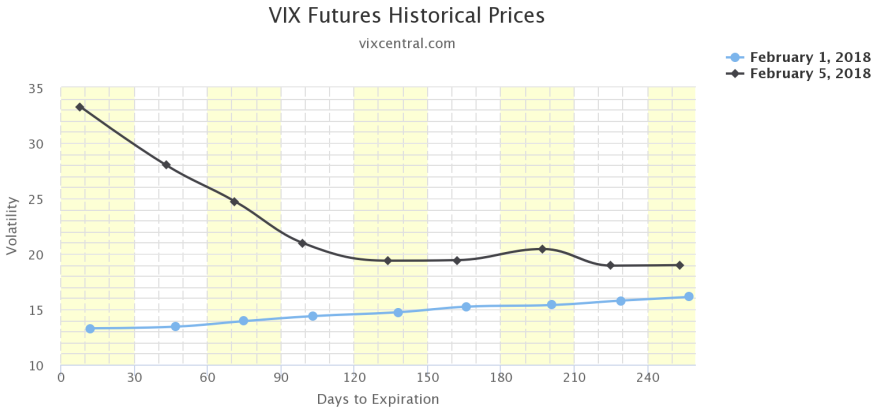
From normal to inverted term structure: Crude oil futures

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



From normal to inverted term structure: VIX futures

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B



Marking-to-market a forward contract

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Suppose you bought forward at time 0 at the forward price F_0.
- Now it is time t and the forward price is F_t.
- Question: What is the current value of your forward position? (We sometimes refer to this as the marked-to-market value of the forward position.)
- The time t value V_t of a long forward position entered into at time 0 is given by

$$V_t = e^{-r(T-t)}(F_t - F_0).$$

– if you sell forward at time t, your payoff at maturity is equal to $(S_T - F_0) + (F_t - S_T) = F_t - F_0$ with certainty;

– the present value of this payoff, $e^{-r(T-t)}(F_t - F_0)$, is the time t value of your initial long position (since the short position at time t is worth zero).

Marking-to-market (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Plugging in the no-arbitrage expression for the forward price F_t yields the marked-to-market value as a function of S_t:
  – e.g. in the case of a non-dividend paying underlying asset, $F_t = e^{r(T-t)}S_t$, and therefore

$$\begin{aligned} V_t &= e^{-r(T-t)}F_t - e^{-r(T-t)}F_0 \\ &= e^{-r(T-t)}e^{r(T-t)}S_t - e^{-r(T-t)}F_0 \\ &= S_t - e^{-r(T-t)}F_0. \end{aligned}$$

- Alternatively, using the FTAP (again assuming a non-dividend paying underlying):

$$\begin{aligned} V_t &= e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[S_T - F_0] \\ &= e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[S_T] - e^{-r(T-t)}F_0 \\ &= S_t - e^{-r(T-t)}F_0. \end{aligned}$$

Remark

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
▷ Forwards
Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Recall that the intrinsic value of an ITM call option is $S_t - e^{-r(T-t)}K$.
- This is the value of a forward contract with forward price K.
- The call is worth more than the forward because of the value of optionality (or "time value").

Futures

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Futures contracts are very similar to forward contracts but there are some differences, which can turn out to be important.
- Whereas forward contracts are typically traded OTC, futures are standardized contracts traded on exchanges - like the Chicago Mercantile Exchange (CME) or the Chicago Board of Trade (CBOT).
- The one specific feature of futures contracts that matters for pricing is periodic settlement.

Description of cash flows for a futures contract

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Like a forward contract, a futures contract expiring at time T is anchored to some variable S_T, typically the price of some underlying asset.
- A long or short futures position on a given underlying, with expiration date T and futures price f_0, is entered at no initial cost at time 0.
- Then it implies a series of cash flows contingent on the realization of the futures price process $\{f_1, f_2, \dots\}$ over the life of the contract.
- We will analyze later how the futures price process $\{f_1, f_2, \dots\}$ is determined.

Description of cash flows (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Specifically, by taking a long position in the futures contract at time 0, an investor agrees to "receive" a series of random cash flows, with cash flow at date $t = 1, \ldots, T-1$ equal to $f_t - f_{t-1}$ and final cash flow $S_T - f_{T-1}$.
  - The final cash flow can be written $f_T - f_{T-1}$ with the convention $f_T = S_T$.
  - Conversely, cash flows to the short party are $f_{t-1} - f_t$.
- Note: A futures position can be closed after each settlement date at no cost.

Example

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- At time 0, you enter a futures contract specifying that you will buy 10 tons of corn at $T = 2$ at price $f_0 = 950$ per ton.
- At $t = 1$, the futures price of corn (for delivery in one period) has increased to $f_1 = 960$: you earn $f_1 - f_0 = 10$ per ton, i.e. $10 \times 10 = 100$.
- At $t = 2 = T$, the market price of a ton of corn is $S_2 = 955$. You have to pay $|S_2 - f_1| = 5$ per ton.
- Remark: You could have closed your futures position at $t = 1$ at no cost.
- Had you entered a long position in a forward contract on 10 tons of corn, maturing at $T = 2$, at the forward price F_0 per ton, you would have "received" a single payoff $10 \times (S_2 - F_0)$ at T.

Uses and misuses of futures contracts

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Futures (like forwards) can be used for hedging or speculative motives.
- Whether you hedge or speculate with futures, periodic settlement of futures contracts makes them risky – if prices move against you even temporarily, it can hurt you badly since you lose money on periodic settlement.
- Nightmare scenario: After losing too much money on futures you are forced to close your position - before it starts becoming profitable again!

Martingale property of the futures price

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Consider a discrete-time setting, with trading and settlement at dates 0, 1, 2, ..., T.
- The interest rate is r_t from t to $t+1$, with continuous compounding, and follows an arbitrary stochastic process.
- **Proposition**: In the absence of arbitrage, the futures price process f is a martingale under $\mathbb{Q}$.
- Proof:[1]

  – Consider a trading strategy wherein you go long in the futures contract at time t and close out your position after settlement at time $t+1$.

  – The cost of this strategy is 0 at time t, and it has a single payoff: $(f_{t+1} - f_t)$ at time $t+1$.

[1] We prove this result only in the discrete-time case, but it holds in continuous time as well.

Martingale property of the futures price (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Using the FTAP[2],

$$0 = E_t^Q[e^{-r_t}(f_{t+1} - f_t)] = e^{-r_t}E_t^Q[f_{t+1} - f_t].$$

- It follows that

$$E_t^Q[f_{t+1} - f_t] = 0 \Rightarrow f_t = E_t^Q[f_{t+1}].$$

- Using the law of iterated expectations[3],

$$f_t = E_t^Q(f_s), \qquad \forall s > t.$$

- Hence f_s is a martingale under Q.

[2]We are using the discrete-time version of the FTAP here. See page 148.
[3]The Law of iterated expectations says that for $t \leq s \leq u$, $E_t[E_s(X_u)] = E_t(X_u)$.

Relationship between futures and forward prices

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- So far we have not addressed the issue of the determination of the futures price process f (for a given underlying and expiration date T).
- What is the no-arbitrage futures price f_t? How does f_t compare to F_t?
- Preliminary observation: In the special case where the interest rate is zero (i.e. the timing of cash flows does not matter), it is easy to see that f_t and F_t must be equal.
  – Indeed, both contracts can be entered at time t at zero initial cost and they pay respectively

Relationship between futures and forward prices

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

Proposition: Suppose r is constant. Then $f_t = F_t$

Proof:

- Using the martingale property of futures prices,

$$f_t = \mathbb{E}_t^Q(f_T) = \mathbb{E}_t^Q(S_T).$$

- Applying risk-neutral pricing to the payoff of a forward,

$$0 = e^{-r(T-t)}\mathbb{E}_t^Q\left(S_T - F_t\right)$$

$$\Rightarrow \quad F_t = \mathbb{E}_t^Q\left(S_T\right)$$

- It follows that

$$f_t = F_t = \mathbb{E}_t^Q\left(S_T\right)$$

Futures and forward prices: General case

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- Now suppose the interest rate follows a stochastic process.
- We consider the continuous-time case. The same argument applies in discrete time (just replace the integral by a sum)
- The futures price is still a martingale, so that

$$\Rightarrow \quad f_t = \mathbb{E}_t^Q (S_T)$$

- The forward price satisfies:

$$0 = \mathbb{E}_t^Q \left[e^{-\int_t^T r_u du} (S_T - F_t) \right].$$

- When interest rates are constant, deterministic or independent of S_T

$$F_t = \mathbb{E}_t^Q (S_T)$$

- Hence

$$f_t = F_t$$

© S.Xiao (2025)

Futures and forward prices: General case (cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Forwards
▷ Futures
Fixed Income
Interest Rate Options
Appendix A
Appendix B

- If the interest rate is stochastic, forward and futures prices are typically not equal.
- The relationship between F_t and f_t depends on the correlation between the interest rate and the price of the underlying.
- It is reasonable to assume that there is a positive correlation between the futures price and the price of the underlying.
- Suppose the interest rate and the price of the underlying are positively correlated, implying that the interest rate and the futures price are also positively correlated.
- Then, on average, at times when you earn money on a long futures position the interest rate is high, which allows you to save at a good rate.
- This makes going long in a futures contract as opposed to a forward contract relatively more appealing, and since both can be done at no cost, this must be reflected in a higher futures price: $f_t > F_t$.

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Appendix A

Outline

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

© S.Xiao (2025)

Breeden-Litzenberger

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
▷ Appendix A
Appendix B

Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

USEFUL FACTS

USEFUL FACTS

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

© S.Xiao (2025)

USEFUL FACTS

Introduction
Binomial and FTAP
Stochastic Calculus
Black-Scholes
Hedging in Black-Scholes
Continuous Time FTAP
Volatility
Exotic Options
Forwards and Futures
Fixed Income
Interest Rate Options
Appendix A
▷ Appendix B

- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

© S.Xiao (2025)